



ECODASH-AQUALINK

SS1

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Git Account Link: <https://github.com/24tebogoR/EcoDash-African-Logistics.git>

Website Link: <https://24tebogor.github.io/EcoDash-African-Logistics/>

Task 1.2 - AquaLink - Project Overview

Firstly, AquaLink aims to provide water access and delivery to rural communities within South Africa. In many rural communities there is a lack of reliable water. In some areas the water may be far away to fetch or there may be unreliable water sources and inadequate infrastructure. In a study on water security in 14 rural villages in Vhmebe District, South Africa, 448 households participated in a study by Malima and Pindihamu (2022). The study found that all rural villages utilized various water sources and that boreholes were used as a significant water source. Additionally, a large amount of time was also spent on water gathering by households. Only 28% of the 38 boreholes in the villages were in use. Water tankers were also used as an emergency water source, but reliability of this source was also an issue. Furthermore, Mpongwana, Shumba and Bracking (2022) also looked at water insecurity among rural households within Limpopo and the Eastern Cape in Goboti and Khubvi. They found water availability among rural households to be unequal and that water insecurity was a problem among households in resource-scarce rural areas. Moreover, Matimolane et al. (2023) have also found water scarcity, inadequate water distribution infrastructure and difficulties in meeting water provision in the rural areas of Limpopo. That said, data from 478 households were used in this study. Many households were found to be using rainwater harvesting as a complement to the available water supply.

AquaLink's - Solution

AquaLink represents a simplified digital model of rural water delivery. The simulation uses an electric water-delivery vehicle that travels through a rural environment while managing its available battery. The vehicle must navigate environmental obstacles such as potholes, fallen trees and rivers while travelling to a Solar Energy Grid to recharge its battery.

The project therefore connects three real-world ideas: rural water delivery, transportation and infrastructure limitations and sustainable energy for electric transport. Also, the simulation does not attempt to reproduce the entire South African water system. Instead, it emphasizes how transportation, energy and environmental conditions can affect the delivery process.

Project Structure

1. Mathematics and Physics Model

Mathematics and Physics has been used to simulate how the AquaLink vehicle moves and responds to its environment.

2. Vehicle Speed and Velocity - horizontal and vertical velocity

```
this.velocityX = 0;
```

```
this.velocityY = 0;
```

Furthermore, its position is updated using its velocity, this allows the vehicle to move across the environment.

```
this.x += this.velocityX;
```

```
this.y += velocityY;
```

3. Trigonometry and Direction

Math.cos() - calculates the horizontal component of the vehicles movement

Math.sin() - calculates the vertical component of the vehicles movement

The vehicles direction is represented using an angle. This allows the simulation to connect the vehicles direction with its movement across the canvas.

4. Friction

The simulation applies friction to the vehicle's velocity:

```
this.velocityX *= 0.98;
```

```
this.velocityY *= 0.98;
```

This gradually reduces the vehicles velocity.

5. Battery Consumption

The electric vehicle consumes battery energy while moving: if (this.velocityX != 0 || this.velocityY != 0) {

 this.battery -= 0.03; This creates an energy management element in the simulation.

6. Solar Energy Recharge

The vehicle can recharge when it enters the Solar Energy Grid/area. The simulation uses rectangle overlap calculations to determine whether

the vehicle has entered the solar area. When the overlap is detected, the battery increases according to the solar recharge rate.

This represents the use of renewable energy to support an electric water-delivery vehicle.

7. Collisions Calculations

AquaLink uses collision detection to detect when the vehicle interacts with the environmental obstacles.

The vehicle and each obstacle have: An X position, A y position, a width and height.

The simulation compares these values to determine whether the rectangle areas overlap.

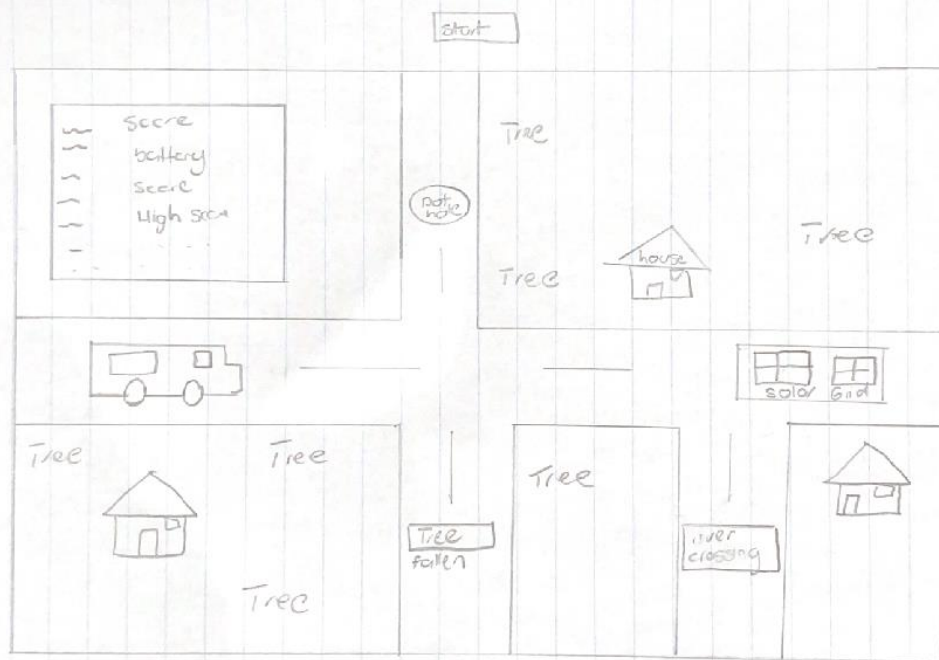
Different obstacles then affect the vehicle differently: a pothole reduces speed speed, a fallen tree can block the route and

a river reduces vehicle movement.

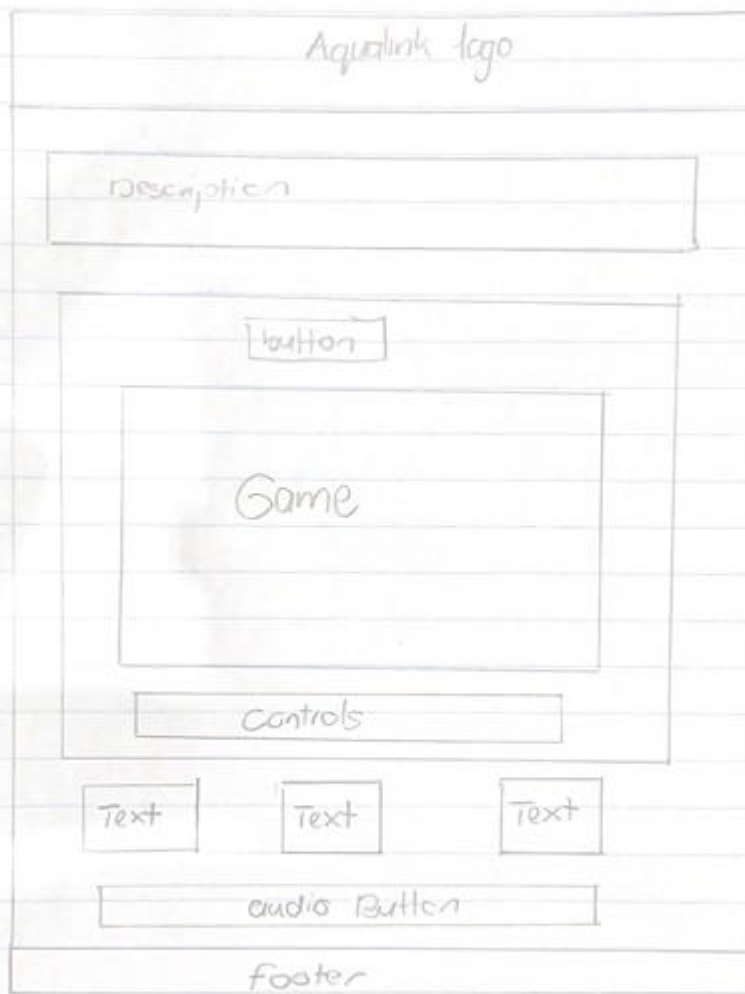
Task 1.3 Wireframes & AI Reflection Log

Wireframes

Wire frame



wire frame



AI LOG

Date	AI Tool Used	Prompt	How AI Was Used	Reflection
2026	ChatGPT	Asked for assistance with JavaScript concepts such	AI explained programming concepts and provided	I studied the examples and adapted the concepts to my own project structure rather

		as classes, constructors, inheritance, methods, getters and setters.	examples that helped me understand how they could be applied.	than using AI as a replacement for learning the concepts.
2026	ChatGPT	Asked for assistance with vehicle movement and collision detection.	AI helped explain how keyboard controls and AABB collision detection could be implemented.	I selected the vehicle movement system, obstacles and route-planning concept and tested the functionality in my project.
2026	ChatGPT	Asked for help improving the HTML and CSS structure.	AI suggested ways to organise the webpage, HUD, controls and responsive layout.	I selected and adapted the layout and styling to fit the AquaLink theme and my assignment requirements.
2026	ChatGPT	Asked for help understanding how battery management and solar charging could work in the simulation.	AI helped me understand how variables and methods could represent battery consumption and solar recharge.	I incorporated the functionality into the simulation and tested how it affected gameplay.
2026	ChatGPT	Asked for help debugging and improving parts of the JavaScript.	AI helped identify possible programming problems and	I tested the suggestions myself and decided which changes were appropriate for my project.

			suggested corrections.	
2026	ChatGPT	Asked for assistance with documentation.	AI helped organise explanations of the development process.	I reviewed and adapted the documentation so that it represented my actual project and development process.

Task 3.2 Reflection Paragraph

I used AI as a supporting and learning resource for the development of AquaLink. I used ChatGPT primarily for learning concepts in JavaScript, planning in-game features, debugging errors and refining my CSS and HTML code.

I utilised AI for learning about topics such as collision detection, battery management, keyboard input, Canvas, object oriented programming, local storage and game mechanics. I also read and tried out the proposed code and then modified it as appropriate to fit my own game.

My own game design choices (such as gameplay, households, solar system, route system, water delivery, rural South Africa, three environmental problems, AquaLink idea and route system) were all created during my planning and implementation phase of the project.

AI was useful for both learning and debugging purposes however I was still ultimately responsible for adaptation, implementation, testing and understanding of the project.

References

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Malima, T.P. & Pindiham, G.K. 2022. Potable Water Security in Rural South Africa: A Case Study of Vhembe District. *African Journal of Governance and Development*, 11(1.2):256–275. <https://doi.org/10.36369/2616-9045/2022/v11si2a3> [Accessed: 11 September 2026].

Mpongwana, Z., Shumba, K. & Bracking, S. 2022. Reflections on Rural Household Water Insecurity: Evidence from Goboti and Khubvi in the Eastern Cape and Limpopo Provinces, South Africa. *African Journal of Inter/Multidisciplinary Studies*, 4(1):217–233. <https://doi.org/10.51415/ajims.v4i1.994> [Accessed: 11 September 2026].

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